CHAPTER IV
LIBRARY AUTOMATION AND NETWORKING

4.1 AUTOMATION

The name of 'Automation' has been obtained from a Greek name 'Automate' which indicate something which has the power of unplanned motion of self movement. Automation when used in a library circumstances to the mechanization or computerization of all library movements. Automation straightforwardly saying is the process of making plenty of works simply with the help of automatic machine. It is used similarly with the mechanization. In automation, the whole field of investigation loyal the design, application of methods and development and techniques for making a process or group of self controlling, of machines self operating, self moving.

Automation pertains to the theory, technique or art of making a machine, advices or purpose more fully automatic. Presently, it is used in co-occurrence of computerization. Computer and information processing hardware play a major role in the process of automation by reason of inherent ability of a computer to growth decisions in impact, govern or control the process from the information collected by the computer concerning the position of the process.

4.2 LIBRARY AUTOMATION

In the current era of information explosion, automation of library services is, consequently, imperative for effective working and efficiency of a library system. The advantageous of automation is rising upon the library scientist not only due to abundant flow of reading material and information but also for apparent viability and economic consideration. There for, economy and efficaciousness of library systems, both, become most important as a result of automation. It could be it is for this reason that Kimber express, "Library is no different from other office or factory considering the use of computer to perform a given job of work most economically, where the different does arise, however, is in the concept of libraries service".

Appropriately, computerization brings about nearly a revolution in the functioning of a library. Computer is very productive not only for management and library operations but also similarly useful for library users. Processing of even huge

quantity of data becomes greatly easier through computer. Traditional method of information dissemination and data processing is now becoming old. This conventional method of working is too time consuming and cumbersome. Hence, from material preservation, cataloguing and data processing-all can be done much easier with the help of computer. There for it ware, a touch of timelessness and availability of material through computers supply an added impetus and energy to the entire library network. While, many use of the information provide into the computer is possible mostly, as it were, in a jiffy. Automation of library facilitates most efficiency and accuracy and library staff find a lot of support of equipment utilize in automation and their regular work performance. So plenty of facilities such as COM, use of database regardless of geographic distance with the help of telephone and telex, satellite, etc., are accessible for uncomplicated operationalization, e.g., On-line Computer library Center database can be used by approximately ten thousand libraries in thirty nine countries. **(Gupta Sangita 1995).**

Library automation covers use of computers and other semi automatic tools. These are semi automatic because human intercession is involved in grater in expand, so when we talk of library automation, in present, it is significantly the use of computers, associated peripheral media i.e., optical media, magnetic disk etc., computer based services and product in library work. In current times, even the related topics such as networking of automated system and information retrieval systems are also regarded as part of library automation, Although computers have a important role to play in library automation, reprographic and telecommunications technology have an same important role because of the extend of support they offer. **(Singh Gurudev 2007).**

4.3 DEFINATION OF LIBRARY AUTOMATION

"Library automation may be defined as the application of automatic and semiautomatic data processing machines (computers) to perform traditional library housekeeping activities such as acquisition, circulation, cataloguing and reference and serials control. Today "Library Automation" is by far the most commonly used terms to describe the mechanization of library activities using the computer". **(Uddin, 2009).**

4.4 PROSPECTS OF LIBRARY AUTOMATION DEVELOPMENT

The recent trends of library automation systems are characterized by several strong changes brought about by technology and economic development. Unless radical change in either technology or the economy occur, these trends seem likely to continue dominant and to determine the course of library automation development for some time to come. Chief among these trends and easily the most conspicuous is the heavy emphasis being placed on library networks that is systems involving more than one library.

The entire programs of the 1973 Clinic on Library Applications of data processing was devoted to networks and the log-awaited draft report of the National Commission on Libraries and Information Science contained as its major recommendation a proposal for a national networks, evolving from the existing local and regional ones. As Bierman has noted the folly of individual libraries proceeding on their own to use develop major automated system solely for their own use seems well recognized and accepted.

Networks come in many styles and flavours. One of the most common is the cooperative center for providing catalogue cards of which the outstanding and most successful example is OCLC, the Ohio College center. Many other state and regional network are now tied into OCLC and dependent on it to a greater or lesser extent for the services they offer. A second type of network is also concerned primarily with technical processing but attempts to provide a means of cooperative acquisitions as well as cataloguing. Successful ones include the BCL program in Massachusetts and the College Bibiocentre program in Ontario, Canada. The mortality rate for this kind of network is high; however as the death of CALBPC and ANYLTS make clear.

A third type of grouping sometimes called a network is merely a loose consortium of libraries with special procedures for expediting inter library loans, often using teletypes. Most of these are not automated in any significant way and so lie outside the sphere of this book but some have been working for years to implement online networks for the same purposes. A universal interest in networks has several manifest causes. For one thing declining library budgets and the starting increase in book and serial cost have made the need for cooperation between libraries more urgent. Resource

sharing has become a byword and the resources implied are both bibliographical and financial.

Online networks enable libraries to know quickly and efficiently which books are in other number of duplicate copies needed. Bibliographical data can be shared more effectively, too as the OCLC system demonstrates. A final and forceful incentive is that the sources of funds for purely local automation projects have almost dried up the Federal grants that were plentiful in the 1960s disappeared in the 1970s foundations have been less willing to fund project unless they were truly innovative and most libraries own budgets are being held static or reduced instead of being increased. Money for cooperative project that can reasonably be expected to reduce operating costs can usually be found but not money for local experimentation.

Several technical developments have also helped to make the network trend possible. The improvement in communication between computer system and system users has already been noted, but it should be pointed out that this extends to communication between libraries as well. Telephone and teletype have long made some inter library communication possible but online networks allow librarians in many different libraries literally to share the same working files and catalogues in real time, almost as if they were working in one of data by several orders of magnitude and the cost of online terminal has declined almost as dramatically, while labour costs have continued to rise. The combined force of all these has made the economic argument for networks compelling.

Work in the development of standards has also facilities networking. Without MARC as a national standard for machine-readable bibliographic data most of the successful networks would not only to MARCs brilliant developers at the Library of Congress but also to the many librarians and other who worked hard to promote widespread acceptance of the standard though professional organizations such as the American Library Association Science and Automation Division Standards Institute and the International Standard Bibliographic Descriptions for monographs and serials opens the way for standard Serial Number offers promise for eventual control what sticky body of data. These and other standards activities have done much to stabilize the future of networks.

Assuming that the trends mentioned above continue the development of library automation in the near future can be predicted with some degree of confidence. For catalogue card production, very few new projects are likely to be choosing to join a network tied to OCLC or a similar, using it for catalogue searching and resource sharing as well as for card production. Attempts to use such networks cooperative acquisitions are less likely to succeed and seem destined to fall into disfavor; Acquisitions work is more is more likely to be one on commercial package system development effort of Columbia, Chicago and Stanford Universities found that acquisition procedures different so widely that a joint effort at systems development in this field was impractical. This situation is unlikely to change in the near future and local increased use of online techniques. **(Tiwari P, 2010)**

4.5 AREA OF LIBRARY AUTOMATION

Automation has revolutionized the libraries nature, concept, approach as well as its away of functioning. Computers are being applied in different library routines. For good management of funds, automation in different acquisition operations is very important. These operations may be ordering, preparation of other slips, generation of reminders to vendors, verification of books with order file and budget control, invoice, maintenance of the accounts of every monetary transactions, print outs of received and non-received documents.

Serial control in libraries is generally extremely complex, due to the availability of serials in different formats and media. In the computer based serial control system, serials are assigned with particular identification codes one of the codes which is assigned to serials on a world-wide basis is six-character alphanumeric CODIN Code. Different routines to such as preparation of orders for subscription, generation of reminders to suppliers for non-receipt of issues, maintenance of records of serial holdings, preparation of renewal titles for subscription, updating the record file, can be automated.

Due to requirement of additional access points and comprehensive search facilities, the necessity for computer based catalogue is realized. An individual library or a group of libraries may undertake the computerization of different processes such as

preparation authority file or subject heading list, shelf list and arrangement of the catalogue according to classified, alphabetical or any other the library decides. Growth of Micrographics has made it possible to renovate the computer printed catalogue into COM catalogue. COM catalogue is economical, compact durable and. Circulation was the earliest library operation to be automated.

The basic features of computer-based circulation system are charging, overdue controls, discharging, reserves, etc. This system should be capable to perform different jobs such as registration of library members, issuance of borrower cards, charging and discharging of documents, reminders of over dues, updating to record files, identification of items on lone, identification of items and due date slips and generating the orders for stolen books automatically, In an online computer based circulation system, can retrieve up-to-date information immediately through the terminal.

Computerized Information Retrieval System exploit information contents in all types of documents to provide a variety of information services to library clientele. The purpose of Information Retrieval System is to make comparison among the representation of enquiries and representation knowledge for example, information entities. The procedure of comparison is commonly referred to as matching. Speed and efficiency are the two most significant criterion of Information Retrieval System. An index is an effective retrieval tool. Computerized indexes can be complied more speedily than their manual counterparts. The format of output from computerized indexes offers better flexibility than do the printed ones.

Selective Dissemination of Information is an alerting information services through which users receive regular notification of new literature in their area of interest. A computer maintaining a machine retrieval bibliographic database selects and prints out the items which are matched to a user’s profile. The whole operation of Selective Dissemination of Information is in five phases, viz. Profile generation, Database generation, Searching database with users profile, sorting the answers and supplying list of bulletins. The Selective Dissemination of Information search has diminished as interactive searching has become readily available at a reasonable cost. Presently most searching is retrospective. Users request is compared with the entire database maintained in the library and a list of matched item is generated.

Thesaurus is a significant tool for effective vocabulary control in computerized Information Retrieval System. Computerized thesaurus in specific fields displays hierarchical relations among the terms. The subject strings obtained by facet analysis and expressed in natural language are manipulated by computer to generate the links for reprinting the relation between the identified concepts. Computers have brought about a revolutionary metamorphosis in the dynamics of library system. It would not be exaggeration to say that ours is an age of computer. The pervasive impact of computers will no doubt emerge, in the years to come, as a main breakthrough in library and information science. **(Gupta Sangita 1995)**

4.6 ORGANISATION OF LIBRARY AUTOMATION

A successful library automation project depends on good management. The manager must know and understand the task to be performed and the individuals involved in the project and must have the confidence of his or her staff and manager. Library automation involves change in the ways an organization conduces itself and represents itself to those to whom it provides services. Like any activity that involves change, planning for it, implementing it, and managing it are not simple tasks. Individual roles within the organization, the organizational structure, and service patterns will change as a consequence of automation. Management faces the same challenges experienced by any newly automated organization. In this case there are two dimensions for management the line responsibility of managing an organization with ongoing service responsibilities and project management of selection or design and implementation of automation system.

A manager can receive valuable assistance in planning and implementing a library automation system from a new key individual in an organization but consultation within a wider group is critical. Any organization considering automation should create a group that represents the constituencies that will manage, fund, use, operate, and maintain the system. Such a group serves many functions and may exist for a long time. The group can provide valuable information on specific requirements for the system, evaluate the comments of others about the necessity and feasibility of certain features and serve as a sounding board for the individual who manages automation project.

The importance of such an advisory group cannot be minimized, because systems are neither selected nor used in a vacuum. The sooner, all can participate in the process, the more likely its success. The duration of such a policy group can vary widely. It might be formed to provide advice on function and features of a proposed system, but could remain active to evaluate alternative designs, vendor proposal, prototype systems, testing, and implementation to give continuing advice on systems operation.

When ant type of computer system is installed in an organization it changes the way operations are performed the task of the staff and the relationship between the library and external units. Staffs that work on the automation project are involved in the day-to-day task of bringing the system once it is operational need to know about it and how it will impact their lives. The way these staff members conduct themselves within the organization and while providing services to users is a complex process which requires thoughtful management. This is especially true when the changes raise the possibility of economic insecurity and fear of the unknown or threaten social relations and habitual practice. The organization must take into account that individuals perceive events around them differently that people are motivated in widely different ways and the attitudes are formed by complex sets of variables and events.

Project team staffs face other issues. They are part of a group whose members have certain roles and status. The group as a whole is accountable for the project, but the members of the group must do the work. Individuals on the team may have personal goals different than the group goals or they may be in complete harmony with the project team. The structure of an organization may change as a result of automation activities. The installation of a new system or change over from one system to another requires skilful management. Departmental structures may change and some departments may disappear.

New organizational units may be formed and new staff functions created. Such changes may be difficult to implement because of organizational inertia. Realignment of departments represents a threat to existing managers, but may present an opportunity to improve performance and service. For example, a streamlined piece of software may render existing organizational task divisions illogical. If boundaries need to be redefined, managers must consider the needs of individuals as well as the goals of the organization.

Patterns of communication between departments may also change. For example, usually the library is not a discrete enough unit to manage its own pool of funds, but relies on another organization to handle the burden of management and disbursement of funds for an acquisition system for instance. The link between departments may also change. For example, usually the library is not a discrete enough unit to manage its own pool of funds but relies on another organization to handle the burden of management and disbursement of funds for an acquisitions system for instance. The link between the library and the unit that manages the bank accounts must be renegotiated to take into account the effect of the new computer system. This link may involve routine exchanges of data, but managerial intervention in the initial stages is crucial to success.

The steps to implementing a library automation system are generally the same whether converting as existing system or introducing an automation system for the first time. Both next steps involve determining the objective of and constraints on the project, an analysis of existing system or processes, design or selection of a new system, testing the system, conversion to the new system. It begins with an assessment of the need for an automated solution or recognition that an existing automated system is inadequate. This assessment uses informal techniques and formal methods. Whatever the method, the result is recognition of the need for change or acceptance of the status quo.

Planning activities are not confined to development of functions and features. Once need is determined and required features have been defined, the manager of the project can decide how to proceed with the help of top level managers the library board, executives of other governmental units, staff and the advisory committee. A timeline is developed, funding obtained, staff recruited, physical space obtained, contracts issued and the system installed, tested, and implemented. Staff must be trained to use it and library users educated in its features if appropriate.

An organization needs a clear statement of what it wants to accomplish in terms of a needs or requirements assessment. If there is a need for a new library automation system, the next planning task is to assess the functions and features that this system should contain. If a new online catalogue is considered what should be the nature of the user interface? Should the catalogue be accessed through computer networks? How many bibliographic records should the catalog be designed to manage? How many terminals

should be attached to it? This type of list is endless. The examples given here are general, but a completed list is much narrower and more specific. Assessing the system requirements, defining them precisely, and testing the acquired systems fulfillment of the needs are critical to a successful automation venture. A statement of objectives includes the desirable function and features of the system such as user, management, hardware and software requirements. A particularly desirable approach to this task is to be guided by what one wants to accomplish, not by what one thinks is feasible. **(Malavya U.C., 1999)**

4.7 TASKS OF LIBRARY AUTOMATION

The steps to applying a library automation system are basically the same whether transforming an existing system or introducing an automation system for the first time. Both are being by planning and problem identification. The upcoming steps involve determining the objective and restrictions on the project, an analysis of processes or existing systems, selection or design of a new system, evaluation of the system, testing a new system, conversion to the new system and staff and user training. It being with an assessment of the requirement for an automated solution or recognition that existing automated system is insufficient. This assessment uses unofficial techniques. Whatever the method, the result is recognition of the requirement for change or acceptance of the status quo.

Planning activities are not sure to development of features and function. Once need is determined and essential features have been defined, the manager of the project can make a decision how proceed, with the help of highest level managers, executives of other governmental units, staff, the advisory committee and the library board,. A timeline is developed, obtained, funding, staff recruited, contracts issued, physical space obtained, and the system tested, installed, and implemented. Staff must be learned to use it and library users educated in its characteristics if appropriate. The work of library automation at following:

1. Objectives

An organization requires a clear statement of what it wants to fulfill in terms of a requirements or needs assessment. It there is require for a new library automation system, the upcoming planning work is to assess the functions and features that this system

should contain. If a recently developed online catalog is considered, Should the catalog be accessed through computer networks? How many bibliographic records should the catalog is designed to manage? How many terminals should be attached to it? Should be the nature of the user interface? This type of list is unlimited. The examples given here are common, but a completed list is smaller narrower and more specific. Assessing the defining them precisely, system requirements, and testing the acquired systems fulfillment of the requirements are critical to a successful automation project. A statement of objectives includes the desirable features and functions of the system such as management, user, software and hardware requirements. Any especially desirable approach to this work is to be guided by what one wants to fulfill, not what one thinks is feasible.

2. Constraints

Next analyze what is practical. Reality sets in during this stage. Is there sufficient time to complete the project? Are there adequate funds to pay for staff, building space, equipment, hardware and software? Is staff with the exact skill mix available? Objectives may have to be modified to meet constraints, if resources are inadequate to meet objective.

3. System analysis design

Specified set of objectives mediated by constraints, the upcoming step is to analysis the operation of the present system to see how it works and to determine its features and functions. The system to be examined is either one currently used in the organization such as a manual circulation function or one proposed by a vendor or an automated circulation system. The required outcome of this phase is a best, wall documented understanding of how the present system works its acceptable features and those requiring redesign.

The resulting document serves as a basis for upcoming phase of the project the design of a new system or the selection of a particular vendors offering. System design is many times long and complex operation that assesses user demands, constraints and existing system operation. It should result in a separatist for a new system that meets individual needs and formal requirements. Since most libraries do not improve their own automation systems, the process of improving an entirely new system is

excluded. However some planed activities are often encountered in configuring vendor systems to meet specific user demands.

Testing is a continual process when a programmer initiates a code, when the staff informally reviews the system, when the product goes through various stages of quality assurance and when other programmers use the cade. Testing continues even on a finalized configured product ready to be hand over by the vendor. Whether a system is designed within the organizations or purchased, it should be implemented on a trial basis specifications developed at the start.

Sometimes preliminary testing of a system configured to the libraries requirements is done in the vendors offices using the libraries data. Occasionally a vendors system is transferred to the library and the system is configured there. The testing process can be really rigorous depending on the requirements of the organization. Both software and hardware must be tested. While there are comparatively standard methods for making sure that hardware is functioning properly the testing of library automation software is not so codified.

There are machineries for testing large online systems which include a computer system being connected to each other computer whose function is to imitate a set of terminals every submitting requests to the machine being tested. The emulator stores system of commands to issue and anticipated responses it logs, date and timestamps all communication with the test machine. Generally testing is not this formal. Many times a group of library staff members are distributed the task of unofficially trying out the function of the system. They log on paper or with screen tip any anomalies that occur. Problems are then communicated to the vendors and designers.

4. Conversion

If the library transform a manual system to a computerized one significant conversion activities are required. For a circulation system every borrowers records are keyed into the automated system new library user cards are issued with barcode Identity (ID) numbers attached, the bibliographic description for all book in the collection is recorded in a computer file a barcode is set up in all book to identify it specially and each bar-coded books is connected to a bibliographic description. Borrower types must be identified and borrowing privileges for each borrower category and materials category

must be established. These works are not trivial and may take 100 of hours for a large library. The results must be nearly error free.

For the time of preparation for automation the manual system should continue operation. When use of the automated system started, backup safeguards must be in place to protect the honesty of transaction. Conversion may look simpler when a library proceeds from one automation system to another but there are disadvantages and advantages to this scenario. The advantage of the staff has experience using an automated system and may learn the underlying concepts. On down side no 2 vendor hardware or software system operates the same way unload or load information in the same format handles every field or structures their files in the similar way. The conversion becomes complicated when it requires informal knowledge of not 1 but two library automation systems. Often it demands knowledge of two hardware platforms and two operating systems as well.

5. Operation

After a stable systems exists, can be placed in operation. The staff should understand the system rigorously and procedures must be written and followed. Among the additional significant activities to be documented are how to activate and stop the system, how to suit it down under urgent situations, how to detect problems, how to make backups of the files and who to notify under a variety of situations.

6. Training

The most of individuals who will operate the system required training. At the beginning management requires to know the common characteristics of the system how it works if it is working good and how it differs from the present system. Management also required periodic status reports on system operation. System managers require training in the hardware on which system will run the operating system and the library automation application.

There is a desire on the part of libraries to shift the responsibility for understanding a library automation system into the consultants, developers and vendors. This is a mistake and can guide to an implementation difficulty when the library thinks the system should perform in a confident way which is different from how the developer and vendor believe it operates. The staff that will operate the system required large

training in its use for all the fact that developers frequently think that the operation of the system they have established is self evident and once in use will be without problem. Normally this is not the case. Therefore the additional documentation obtainable in written form in videos and in assistance programs integral into the system from vendor technical support workers and from human beings serving as instructors the grater is the likelihood that the workers will receive and use the system successfully.

7. Evaluation

Once a system installed and has been operating for a sensible period of time inferences can be made about its capacity, reliability and performances. The majority library automation systems generated reports such as the overdue notice sent, number of orders processed, serial titles checked in and so forth. But other measurements give significant information about the internal operation of the software itself, such as percentage of space on the system disk drives that is filled, response time for online requests, disk input/ output operations performed by time of day, memory utilization, central processor capacity utilization by time of day, and communication line utilization and error rates.

An additional qualitative form of evaluation should also be take on that involves comparing the objectives established for the system with result from its operation. This evaluation takes place at least two times, once when the library is about to receive delivery of the system and again after the system has been starting. Most library managers will select to purchase rather than built their individual library automation system. Since a broad variety of systems is available on the market for large libraries, medium, small and tiny, only libraries that are extremely large, have special requirements, or have the funds and interest are likely to pursue designing and implementing their individual systems.

Even so don’t stop reading this book now. The more that is noted about how these systems works, the greater the probability of choosing the best system and implementing it successfully. Of most important concern is the quality of the product and how well it meets libraries needs. This can be ascertained by the libraries individual evaluation of the system and by making contact with present and past customers of the vendor. User groups can supply a forum for interchange of thoughts, sometimes apart

from the umbrella of the vendor. Cost factors present another concern the porches price of the software and hardware, conversion costs, maintenance costs for both, cost of additional training, and type of training and customer support provided. These entire elements will be incorporated into a contractual agreement between the vendor and the library that will include a schedule for purchase, installation, and testing. Prior to signing any contract, library officials should look into the financial stability of the vendor, since a considerable deal of time and effort will be expended in converting the present system to the new environment no library wants to goes insolvent of ceases involuntarily from a vendor who goes insolvent or ceases operation. **(Gopal Krishna 2005).**

4.8 TECHNIQUES OF LIBRARY AUTOMATION

The connection between the library and the computer is a changing one. Computer science has been changing quickly ever since its origins twenty years ago and now the library itself are beginning to be changed under the effect of the machine. In changing world librarians have a duty not only to be aware about the ways in which libraries can utilize the techniques of automation but also to know of the changes automation can carry to library services in the near and maybe the more distant future. It is uncomplicated in this 20th century to strive for change easy for its own sake just as it is even easier to attempt for retaining the status quo.

Those librarians whose work it is to decide for or against the specific change of automation require especially being on their guard against these 2 attitudes. They require making their decisions in the light of facts, facts about their libraries as they are at the movement what services are provided to what general public by want library staff and at what price and similar facts about any present system of library automation. In this library is no changed from any other factory or office considering the use of a computer to work a given job of work most economically. Where the dissimilar does arise however is in the idea of libraries services.

A company may change a certain price for its artifacts and so confident make an overall profit on its operations. In such circumstances a computer can be display to act either to increase or to lessen profits. But there are hardly any libraries today which leave in order to make a profit for their proprietor and few are essential to balance an income

from subscription against their charges. How then can a price be fixed against the services which a library can give and how can this price to be calculated so that automation may be seen to be explained? An answer to those questions can be found in 2 ways. Firstly anyone industrial librarian noted that this job is to fulfill his company's information requirements.

He a specialist in information task can supply information to company employees extra effectively than they can obtain it themselves and so that importance of his libraries services can be expressed in phrase of hours and minutes of saved time and hence in financial terms. An analogous quarrel can be used of the reader services function in university and public libraries. Secondly the advantage of libraries services can also communicate quantitatively in terms of the chief librarians bargaining authority at meetings of the library committee and so in terms of the difficult cash vote to him to authorize him to run his library.

It is easy therefore although hard to express the value of automated library systems in quantitative terms and so to contrast an automated library with its conventional equivalent. The resolution to automate or not to automate can then be taken in the full light of the reality. That is decision seems frequently to have been taken in incomprehension of the realities does not invalidate the arguments kernel services merely to throw into clearer reassurance the well known truths that librarians are as a busy, class that libraries try hard for enough money to buy every the books they want let alone finance a actual automation feasibility study and that it is plenty of easier to obtain money to purchase computer equipment than to purchase books.

The final situation then in majority cases will be that developed library services have been bought at a price. It the librarian has been capable to justify this to have committee then as per my argument the automated system has in case broken even. We should look a little more closely into the reasons why librarians strongly consider utilize a computer in his library in order to make better its working in some way. And the first kind of cause which we shall term emotional is one regrettable outcome of the automation revolution through which the developed countries of the world are now passing. Because as per the emotional argument computers are efficient and modern a library which utilize a computer in some way must necessarily also be competent,

forward looking and almost certainly therefore a better library. It is really true to say that those libraries which have started computer systems are forward looking and modern.

The library automated systems may well be efficient and the library may be a better one. But it is surely not necessarily true to say as the emotive argument does that if library users a computer then it well be all these activities. That this reason exists at all even if only as an view of mind is a direct result of the false attraction ascribed to the computer and the superficial thinking and writing of those accountable for automating aspects of library procedure in the future now is really the time for some libraries to analysis with ways and means of allowing computers to take their actual place in library work.

These analyses should yield both quantitative and qualitative results. So far there have been too plenty of the former and too hardly any of the latter kind. A computer is an electronic machine for operating symbols. Enemy about purpose we can readily make this definition too meaningful if we state that a computer can be regarded in plenty of ways as being similar a clerk. And if it is a clerk then we may be capable to use it to do clerical work in libraries more efficiently, more quickly and accurately than it is done at present. It might even be possible to instruct it some intellectual or professional jobs. Being a machine it should not get tired under hard work loads and in fact should become plenty of efficient as the work load growth so that as library increase this can be accommodated without corresponding increases in staff.

From what we aware about computer application in other areas we should expect automation to be practicable only in big libraries or in cases where a number of little libraries amalgamate together to form one big unit for processing purpose. However we know also that computer time is developing steadily chapter as never plenty of advanced machine are belittle whereas the time of human clerks is becoming steadily plenty of expensive. So we forecast that automation will slowly become feasible in smaller libraries. This tendency will be accentuated somewhat by the appearance of cheap, very small computers but it must be remembered that plenty of library applications required a big machine. The development of online techniques which are especially suited to some sections of library work will make it practical for libraries to expect to

have online access to a powerful and big machine rather than consider purchasing a small machine for their individual use.

A satisfactorily set of procedures exists leading from the point we have at the moment reached that of having general and specified terms the benefits which might be expected to accrue from using a computer to that of installation, checking and running an automated system in a library. To follow all these procedures constantly might seem pedantic perhaps unnecessary in lot of circumstances but it is mandatory if the job is to be done honestly, cultivating reason, working in quantitative terms and avoiding emotion. Because procedures will be listed and discussed in one specific way it does not mean that this is the just way or that different amalgamation and distributions of these procedures might not result in extremely dissimilar patterns of overall working only that the fundamental ideas must follow one another in this way.

The first step in the direction of introducing an automated system is to conduct a feasibility study. This may be complete at any level of detail but the reason is to find out whether or not a computer can be used to do a specific job more efficiently than heretofore. It can be an extensive stage and may be broken down into2 parts the first one to find out if the feasibility study itself is value undertaking. It is understanding that may such studies have seen that a computer was not shelf that may such studies have seen that a computer was not really necessary at all and that the information obtained from the possible study was sufficient to develop the previous system so as to make it more competent than the automated one first contemplated. It is during the possible study that systems analysis takes place.

The system be it the entire library itself or just one department is analyzed its technique of working is systematically taken apart bit by bit so that in and all the bit sare known and understood as are the ways in which they suitable together to make the complete. If only one department of the library is being investigation in the relationship between it and the different departments and the library it must be known and understood. Calculation will also take place so that the concluding result of the study can be in quantitative terms. Numbers will be calculated the numbers of items catalogued, bought, lost, damaged, circulated and the way be expected to modifications in the future will be discovered.

As well counted will be the number of people necessary to do the different jobs in the system and the time to do every one jobs in the system and the time to do every one job will be measured. It is very significant that the ends results of all this library movement be estimated or calculated. This can be not easy difficult because it is difficult to apply a measure to a book, student reading, chemist pleased with a mouthful of information and a housewife experimenting with new and different dishes described in a cookery book borrowed from the near local public library. It should be attempted however because any library automated system as well have seen is possible to change the pattern of service as the number of books purchase each week or the time it takes to discharge an overdue book. From all these calculations the costs of the existing system will be calculated. They will nearly certainly be expressed as unit costs the cost of cataloguing the average book or answering the standard reference question.

With each other details of the services performed by the existing system and in particular any problems and difficulties associated with it any appearance which are like to become disappointing in the light of future developments will be suggested and all will be used by management to make an educated resolution about the introduction of automation.

By the reason of the introduction of computerized working is not the only and not necessarily the better solution to the problems that will have been brought to light the next step of the study is just to find the better solution automated or not automated. It is very desirable therefore that the study must be carried out by persons who have no directed attentive in seeing that a computer system is installed it may be even using a specific maker of computer. They must be perfectly free to advice on the optimum course of action. Like sage the investigators should have considerable experience of the same studies and be will informed on all tasks study techniques and data processing.

Needless to say there is a considerable shortage of such people. Feasibility studies therefore are not inexpensive and so are often omitted and skimped entirely. If the introduction of automation is advisable and this is the only case that actually concerns us here then the second part of the study will involve the choice of a computer and the different necessary ancillary items of stationery and equipment required by the system. There are appreciable differences between the different metals and makes of computer on

the market so that the choice of the machine almost all suited to the purpose on hand will have a market impact on the final operating ability of the system.

The conclusion result of this second part of the study is a broad system specification. It will indicate the machinery or hardware required and in common terms how everything will work. The all over work load handled will be given as will the approximate costs. Manpower requirements will almost definitely be different from those of the former system for keypunchers will put back clerks and it will be required to have a senior staff member with data processing knowledge in charge of operations and one or two programmes may be essential on a permanent basis.

Staff requirements will be extremely different where a library installs a computer for its individual use from the case where it only uses a short time each week or somewhere else machine. Because computers work in ways completely different from human beings the movement of work in an automated system can contrast with quite considerably from that in the commensurate manual system. Old departmental problems can therefore be broken down of the introduction of automation as same processes in different departments are combined and as repetition of work is eliminated. The inclusive organizational structures of the library can be significantly altered and this will be appreciated and stated in the practicable study.

The automated system is considered in any detail it is clear that it offers a range of feasible products and services considerable winter than doe's orthodox system. The compilation of all types of statistics is but one example of this. A library can anticipate therefore to enlarge the scope of its operations by introducing mechanized routines and it is attractive either that the them doing the study include a better knowledge of library staff tasks closely with the them in order that possibilities of winter activity are realized.

From the concluding report of the feasibility study it should be a comparatively simple issue for the library management to make a decision on a course of action. Only in the big libraries would a feasibility study be likely to specify clear overall cast diminutions through introducing automated methods. In other libraries the position would almost certainly be that at the moment a computer aided system would cost increased but would bring additional benefits. It is up to the management to plan whether these benefits

justify the rise costs bearing in mind that in time costs will tend to change in be to the advantage of automation.

The method for implementing an automated circulation system e.g. will be different from those essential by a system embracing everyone library technical processing routines. In a university situation the library would require to make use of a central university computing facility more than purchase its individual machine or use the services of a business bureau. This will affect not only the kind of machine used which is likely to be best suited to mathematical than to library task and the languages programmes are written in because programming may well be finished in the computing laboratory it will assume the staffing need in the library.

The general computing policy of the organization of which anyone library is a part will also assume such factors as e.g. the frequency with which computer runs can be made, is planned to be available, whether online working is, and so be an significant factor in the design of the library system. In big organization it frequently that a library in another division and has already automated some of its work. If may be possible to take these programmes over just as they stand or with only little modifications. **(Verama Kusum 2008).**

4.9 PUBLIC LIBRARY AUTOMATION

Some libraries have adopted automation of basic library functions such as acquisitions, cataloguing, circulation and access to OPAC. Some libraries are also using computers for related activities for example:

1) Stock control and stock taking.
2) Accounting, community information and book selection.
3) Book reservations but there are libraries which have not automated any of their library functions.

Some libraries have limited use of computers. The use of Word Perfect restricts access to online operations. Database management system would be useful for online library catalogue. Some of the public libraries have not automated any of their functions. They have a plan to automate in future. Some libraries think their present manual systems are adequate and automation would not bring any improvement in services. Also they

have no continuous power supply which is required for the operations. Also funds are not available to meet the high cost of initial software and hardware and further up gradations.

The most of the Automated Libraries services have their automated functions for branch, regional, district or affiliated libraries. They have also provided service points at various places. Most of the libraries are using automated systems for most library functions. But because of lack of funds, automation was not possible everywhere. Sometimes lack of experts, inadequate support or infrastructure like buildings, electricity etc. restricts automation.

4.10 HISTORY OF LIBRARY AUTOMATION

Historically, the word automation was used more valuable to indicate process control. In the beginning, library automation meant only mechanizing some of the more respective work such as periodicals and circulation control. The concept of automation in our society was the brainchild of a talented librarian, Herman Hollerith, staff in the Census Bureau of the United States, who invited the punched card machine.

The concept of inventing the punched card machine came from Dr. John S. B., Director of the Surgeon General's library (US National Library of Medicine). Dr. John S. B., recommended Hollerith "there ought to be a machine for doing the entirely mechanical task of tabulation and similar statistics". In year1890, Hollerith invented a machine using punched for tabulating census data on the basis of the recommendation from Dr. John S. B. In year1896, Hollerith established the Tabulating Machine Company, which later becomes the famous International Business Machine (IBM). This example indicates the involvement of a librarian in the automation procedure and it become very important in initiating a trend. **(Asari K. Ravindra 2009).**

4.11 ADVANTAGES OF LIBRARY AUTOMATION

Library automation by using computer application in the library and information function and services has the following advantages

1. Speed - Information processing is done faster which improves work flow through the library.

2. Accuracy - The degree of precision and accuracy in processing the information is 100% percent. However, it is dependent on the accuracy of information fed to the system.
3. Cost effectiveness – Operating costs are reduced as less man power is required. It can be further reduced if the system is well designed and well managed.
4. Reduction in library workloads – Library workload can be reduced as the computer can do vast amount of work in a fraction of second.
5. Improved services to users – Better quality performance is possible through the use of computers. No delay in processing.
6. Eliminates duplication of work - Duplication of work is avoided
7. Easier access to external database. The data is easily accessible.
8. Access and search of information- Online access and search of information is possible. **(Sampath Kumar G. K 2012).**

4.12 DISADVANTAGES OF LIBRSRY AUTOMATION

Even though you may have doubtless about automating your library, it is significant that you be aware of the drawback library automation may bear. These include the Following:

1. It is time consuming, planning, selecting, and implementing an automated system need important, long-term commitment of staff time. Once choose and implemented, an automated system must be keep in good condition on a regular basis. Having the automated system networked to libraries Local Area Network (LAN) adds additional demands on the media specialists and information professional's time.
2. It is costly, start-up costs, hardware, software, writing, network cabling, ongoing barcode labels, expenses such as supplies for printers, furniture, annual maintenance, technical support and conversion of libraries shelf list into a Machine Retable Catalogue (MARC) may be more than various media centers and small libraries can provide.
3. The requirement of the automated system may not leave staff sufficient time to provide up to date services or to work with teachers, students and other clients. It fact, automation eliminates some work but generates new ones. End user training, ongoing

troubleshooting of software and hardware, and database maintenance place requirement on the information professional and media specialist.

4. Access to the automated system is unavailable for the time of system downtime. This will hamper user access to the collection, mainly if the card catalog or the shelf list no longer exists in the library and media center.

Awareness of the advantages and pitfalls of library automation will help you good prepare for the changes in your task duties. Library automation, like technology, is costly in terms of money and time, and frustration and agony are typical symptoms of techno stress.

4.13 BENEFITS OF LIBRARY AUTOMATION

The Web is building library automation the convention for all types of libraries. Any library or media center that is seeking to start a presence on the Web must have its catalog automated. What is most obvious about automation is that it develops library services and increases accuracy, efficiency and productivity in performing a variety of library operations.

Some added benefits of library automation are follows:

1. It allows patrons to utilize search strategies that beyond those that can be used with card catalogs. Card catalogs can be shared only by subject, title and author Online Public Access Catalogue (OPACs) can be accessed by subject, title and author and keyword, In addition, users can expand their search by using Boolean operation (AND, OR, NOT) and by combining search strategies for example; subject and author, title and author. In addition, OPAC users may restrict their search result by such features as publication date, type of material for example; video, book, and magazine, reading level and language, and they can sort bibliographies by publication date, author, and title.
2. The windows based OPACs permit for hyperlink searching, a new feature that was not possible in character base system for example; Disk Operating System (DOS). Through a hyperlink search a user can find connected records in the automated systems database under a subject or word. The user can also locate related resources that appear on the Web via the Machine Retable Catalogue (MARC) field 856 when

this field is activated. Author search feature that was not feasible in character based system is the optical search. An OPAC that has a Graphical User Interface (GUI) capability permit clicking on icons that represent function instead of clicking on command buttons only. The visible representation of search functions is really attractive to young children mostly, because visual interfaces that are based on pictures or icons are generally augmented with colors and uncomplicated to read text.

3. It allows users to search the library's collection from location outside the libraries walls, users who are equipped with a computer and a modem can dial into the OPAC from remote location, an office and home. Majority automation software is compatible with the Z39.50 standard. Having these standard permitted users to search OPACs on the web using search features and common interface. This means that concerning of the automation software, operating system for example; Windows NT, Windows, UNIX or computer platform users have, they can search these OPACs using usual search interface.
4. It gives users with timely access to library collection can be placed on shelves as soon as items are processed and MARC data are downloaded into a database. When MARC data are purchased with material orders, they remove require for time consuming original cataloging.
5. It supports new means of information retrieval by introducing users to global information. The success and popularity of OPACs make them perfect to coexist with CD-ROM database, the Web, and other information system a on a libraries computer. A Z39.50 acquiescent OPAC allows users to search Z39.50 acquiescent database using the search syntax of the OPAC, thereby eliminating require learning all databases search syntax.
6. It eliminates routine work or performs them more effectively. The circulation function, with include overdue notices, check-in, check-out, and inventory, repetitive, is tedious, and time consuming. Automating these functions can save excellent amount of time.
7. It simplifies and expedites in inventory of library collection. The automated inventory is performed by scanning all items barcode using a hand head device, downloading scanned documents into the automated system, and generating a variation of

customized reports. In a non automated condition, this procedure involves checking a shelf list card against the specific item on the shelf, flagging the shelf list cards for misplaced items, and generating inventory reports manually. The collection inventory that works 2 months to complete in a non automated environment may take 2 week in on automated environment.

8. It encourages collaborative collection development and resource sharing for example; inter library lone. Automated libraries and media centers can develop a union catalogue and join bibliographic consortia and utilities. A user who does not find an book or item of interest in the libraries local OPAC plenty of identity the libraries in the consortia or union catalog that have it. The user can then borrow the item through Inter Library Lone (ILL) or by checking in out from a designated library.
9. It enables libraries and media centers to develop and export MARC records. Records obtained from book suppliers or other sources on disk are imported into an automated system to save cataloguing time. Records can be exported from one system and imported into a recently develop automated system without incurring new cost for retrospective conversion. Exporting records identification essential for relocating from one automated system to another.
10. It decrease the amount of time spend on budget administration, record keeping, serials management and material acquisition.
11. It inspires users, equips them with problem solving, information retrieval skill, and provides them with long-lasting learning experience. In addition, reinforces a positive view about the library and upgrade the image of the information professional or library specialist. Patrons view the library or media center as an essential place for gaining access to global information and consider information professionals or media specialist a powerful information provider.
12. It permits for cataloguing Internet resources and for upgrades them into a local system. Automated system also includes tag 856 to link Uniform Resource locator (URLs) to MARC 21 records, although these features enhance a library’s collection or media centers, they do present some problems. Internet resources are stateless and volatile, so they need validation on a regular basis. If such validation is performed in house, it becomes differ time consuming. Various automation suppliers or vendors

have provided solution to this quandary. One of these is to subscribe to an off line URL service to obtain acceptable URLs on CD-ROM arrives, some of the URLs services. A suppliers or vendor hosts the URLs on a server and updates them always. Every URL that is activated in libraries local automated system passes through the supplier's server of URLs for validation before it is requested from the Internet. This service is more advanced than the offline services, but may be more costly.

13. It can be used in collection mapping. Plenty of automated systems have the ability to create collection maps to use for collection growth. In consortia, creating such maps manually becomes very boring and time consuming.

4.14 BARRIERS OF LIBRARY AUTOMATION

1) Fear of adverse impact on employment

When we look into the various library activities like books acquisition, technical processing, circulation and reference services we understand that human interference is required everywhere. But manpower is required for cataloguing. The data which is entered at the time of ordering can be used for cataloguing with some changes. It will eliminate multiple card preparation and subsequent filling. This saved manpower can be utilized for analytical cataloguing or introducing new services. This way there will be no effect on employees.

2) Apprehension that the technology could be too expensive

It is an understanding that hardware and software would be expensive and unaffordable. Actually cataloguing system is the most important which forms the base for other library activities. For this purpose UNESCO has developed PC based software CDS / ISIS and is available at a very low price in developing countries. It works on IBM compatible PCs or UNIX platform. WINDOWS version is also available. The data can be exported in ISO 2709 format without difficulty. INFLIBNET has also developed a public domain library software ILMS which is suitable for DOS and UNIX platform. Lately the PCs and other accessories like printers etc. have become cheaper and so they are affordable.

3) The library staff has to undergo extensive training

The training for the operating the computer system and the programme is required which is given by the software developers but training for CDS/ISIS is available at INSDOC, INFLIBNET and DRTC. For cataloguing a minimum training for one or two weeks will be enough for the librarians. The user can transfer the same data to network environment. If there is a systems department with computer professionals maintenance becomes easy.

4) Lack of support from the management, may be owing to budget constraints

If the management support is not enough then it may be the barrier for development of the library. In this situation librarian has to take more efforts in convincing the management that users will be benefited by the automation. Also the initiative and communication skill play a major role in convincing the management to get more attention.

5) Retrospective conversion of data

The manpower saved by computerized automation can be utilized for conversion and also for analytical cataloguing. This can be achieved in a time bound project. **(TiwariPurushotham, 2010).**

4.15 CONTROL OF LBRARY AUTOMATION

Library automation has to be consistent with standards and guidelines. Human performance inconsistency should be channeled and therefore it is necessary to allow structural guidelines. Individual professionals must have a clear sense of job identity, encouraging professionalism. It should guard against the potential of diffused work. However the exchange of views and thoughts should be promoted among the professionals regarding specifications of the automation process to improve the quality of process and performance of services. This should be guarded against any loss of overall library objectives.

Scheduling of work must be simpler and specific application time should not be wasted in solving problems of communication between professionals. Such good objective specification process would encourage development of specialists knowledge. An erosion of tight boundaries help in analysis and flexible performance of jobs.

4.16 LIBRARY AUTOMATION STEPS

Planning is time consuming, but it is normally cost effective because that time spend planning bring down the amount of time essential for system implementation. Library automaton steps are flows:

Step 1 Describing existing services and function provided by the library

i. Identifying existing technology and utilize in the library.
ii. Identifying existing function and services provided by the library.
iii. Collection and administration basis statistical data

Step 2 Assessing needs and setting priorities

i. Needs assessment.
ii. Who should be participated in planning?
iii. Establish a preliminary budget.
iv. Setting priorities.

Step 3 Translating needs and priorities into specifications

i. Designation specifications
ii. Formulate and distributing the request for professional

Step 4 Evaluation proposals and selecting a system

i. Making the primary cut
ii. Observing system demonstration
iii. Analyzing supplier or vendor responses
iv. Budget
v. Obtaining response from supplier or vendor clients
vi. Making the last cut

Step 5 Putting your system into place

i. Agreement negotiations
ii. Software and hardware installation
iii. Training and practice

Step 6 Retrospective conversion and bar-coding.

4.17 IMPACT OF LIBRARY AUTOMATION

Automation has affected most library processes to some extent, and in many cases has influenced the development of library policy. In the area of cataloguing the effects of library automation have been so far-reaching and profound that a separate discussion of their impact seems warranted. The effects may be seen in four general areas.

1) Extent of cataloguing information presented to the users.

2) Choice of information given.

3) Arrangement of information.

4) Typographical presentation of such information.

The last of these areas is one of the most interesting from a mechanical point of view, but certainly the least significant so far as cataloguing policy is concerned. Libraries, however, have bemoaned the uppercase mentality of computers and related machine since their use in library work began. The resulting catalogues were certainly less attractive and less readable than the printed or typed cards with which most people were familiar.

Changes in the extent of cataloguing information presented to the user under each entry have been mainly deletions rather than additions and such changes are most frequently seen in book catalogue projects. Virtually every descriptive element has been eliminated or abbreviated in one project or another. The Boeing SLIP catalogue restrict authors forenames to one and represents the other by initials other catalogues give only initials with no forenames.

Collection is perhaps the most frequently abridged set of elements. Stanford and Florida Atlantic give pagination only and many catalogues give no collection at all. Notes and tracings are also frequently omitted with the exception of series and contents notes in a few cases. The ONULP catalogues, however give full tracing under entries, as does the National Library of medicine Current Catalogue, the latter because other medical libraries use the catalogue as a source of cataloguing information.

In many catalogues the amount and type of abridgement varies according to the type of entry involved. Normally the main entry is reasonable full and most abbreviation or abridgement takes place in added entries. In the Los Angeles County catalogue however the subject entries are more complete than the author entries, because the subject

entries include an annotation. The practice of abridging entries has been justified on several grounds.

Undoubtedly it began especially in catalogues produced with unit record equipment such as the University of Rochester listing, because of the eight-column limitation of punched card and the printing limitations of tabulators. As Weinstein and George have noted, the one-to-one punched card/printed line relationship and line-length/field-length compromises, all encourage brevity of entry. At least in part however, the elimination of data elements also reflects a desire to cut costs. Simonton has viewed it as a reflection of the traditional policy of many of these libraries of shortening the bibliographical record for more economical processing.

The choice of main entry has long been a critical issue in cataloguing policy and in the development of cataloguing rules, but with the advent of machine-produced book catalogues the need for the concept of main entry may be questioned. When catalogue cards were printed or typed by hand it made economic sense to produce only one card with the data arranged in a standard format under a main entry and then to reproduce other cards as needed typing only the added headings required after the unit cards had been duplicated. Computers, however, can rearrange data in any fashion deemed appropriate, so this rationale is no longer applicable. Despite this fact many book catalogue continue to follow the unit card and main entry concepts.

The ONULP catalogues. For example, and then repeat the title a practice that would appear to have no justification except for mechanically reproduced catalogue cards. A few books, catalogue projects have abandoned this anachronistic practice and reproduce entries under various headings without any attempt to designate one as the main entry even though the entries under each heading are still unit entries. Abandonment of the main entry concept, it should be pointed out, also means abandoning and attempt to designate the person or body primarily responsible for the intellectual content of a work but clearly it makes the choice of entry less critical, particularly in difficult case such as conference proceedings, where the choice between conference title, sponsoring body or editor may be little more than arbitrary.

One of the greatest effects of automation on cataloguing practice to data has been in the arrangement of entries in relation to each other. Virtually all book catalogues are

divided catalogues, abandoning the so-called dictionary arrangement that interfiles all types of entries in a single sequence. The type of division is not consistent from one catalogue to another. Some have separate author, title and subject sections; some combine author and title entries separate. Regardless of the method of division, however, two principal reasons are usually given for the arrangement; users make different uses of subject entries as opposed to author and title entries, according to a number of user studies and the filing or sequencing of entries can be simplified if all types of entries are not interfiled.

The latter reason is normally the more urgent when automation is involved. Catalogues can be divided without being automated and if ease of use were the only justification for division there would be no reason to attribute the popularity of the practice to automation. Filing however, is another matter and simplification of filing for automation purpose is undoubtedly the main reason for the predominance of the practice. Computers can sequence and data with extreme speed and accuracy if given explicit rules for doing so, but traditional library filing arrangement are a mixture of alphabetical, chronological and other sequences presumed to be logical, with numerous general and specific exceptions. In addition to the complexity of the rules, many of them of based on the ability of human filers to recognize information that is only implicit, not explicitly stated and computers can work only with explicit data. At least five categories of difficulty may be identified:

1) Information in the headings must be ignored or removed in order for computers to sort it acceptable.
2) Information that is implicit must be made explicit.
3) Information in headings must be rearranged in order for machines top sort it into the desired sequence.
4) Exceptions must be made to alphabetic order. Some rules require that entries beginning with the same word or phrase be sub-arranged not alphabetically by the next filing element but by bibliographic function or according to the type of entity the heading describe, then places, then things.
5) Punctuation is even a problem because of complications introduced both by the machines and by filing rules. Computers regard punctuation marks as character just as

letters and numerals and sort them into a certain order unless programmed to do otherwise.

In ideal solution to these various problems is unclear but attempts to solve them in automated catalogues have produced a significant amount of change as compared with manually filed catalogues. Most projects have used a combination of four basic approaches:

1) Continue to follow present rules, using people to assign additional codes to headings as necessary to allow computers to arrange the headings as desired.
2) Abandon present filing rules and use the arrangements produced by the computers sorting sequences with a minimum amount of change.
3) Change the headings themselves as necessary so that computers can arrange them in accordance with very simple filing rules.
4) Attempt to program computers so that they will arrange headings automatically, either by present filing rules or by modifications of them. **(Tiwari P, 2010)**

4.18 LIBRARY AUTOMATION IN INDIA

1) Indian National Scientific Documentation Center (INSDOC)

In India some information centers and libraries made efforts to automate their information services and library routines in the 1990's with punch card and with a plenty of second generation purpose computers that were available at Bombay and Kanpur. INSDOC was the head in experimenting with computers for their application in information work and documentation in 1964. Initially they made use of the International Business Machines (IBM) 1620 model I that was available at IIT Kanpur. The first try was with work on data collected for union catalogue of scientific serial. The alphabetization program was started and the punch card out was printed on a tabulating machine.

2) The Indian Institute of Management, Calcutta

The IIM, Calcutta using IBM 029 statistical machine having a print unit started a system for producing a bibliographies and documentation list in 1968. After selecting the title from the journal, it was categorize, the data sheet was prepared, information was arranged in punch cards by key punching and information was recovered through sorter.

3) Indian Statistical Institute, Calcutta

Indian Statistical Institute, Calcutta Show and Holder in their paper on "Union List of Periodicals" gave exact description of the compilation of the Union Catalogue of Calcutta libraries using punch card system at the Indian Statistical of Petroleum (IIP). The IIP, Dehradun making or developed a punch card method for circulation control in 1969-70, all the relevant data ware records on one punch card as per to a prescribed code.

4) Documentation Research and Training Center (DRTC)

At the DRTC seminar held in Bangalore in 1964, Harold Borko of the System Development Corporation, California submitted paper titled 'Experimental Studies in Automated Document Classification' where he presented his studies in the use factor study a mathematical technique for obtaining classification categories for a set of documents. At first Ranganathan felt that the computer created raise in the library profession. However, he came out a small later with the design of a 'document finding system' in which the orthodox catalogue could be searched in response to a query. Harold Borko also recommends a second system, the 'Doc-Finder' a completely integrated system.

Neelameghan published a paper on "Design of Document Finding System: General Features" in year 1968 in this paper Neelameghan mentioned the objective of a series of experiments undertaken in the DRTC on the use of 'Doc-Finder', for example; the computer in document finding. Ravichandra Rao mentioned in this paper 'Automated System for Acquisition and Circulation' examine in detail the Acquisition System giving the different function to be performed by an automated acquisition systems different reports to be generated from order files, and the added function that are desired to be performed by the system.

Ravichandra Rao in this paper titled 'Use of Computer and Sampling techniques for stock verification' mentioned a method of stock verification using a computer readable catalogue of library. In his two articles, Ravichandra Rao mentioned both off-line and on-line circulation system, the information about the documents can be considered either as transitory i.e., recovered in the calculation file only when an item is removed from its normal location or as present for example; stored in the circulation file

irrespective of where the book is. These 2 systems are also known as Inventory Systems and Absence System.

5) Tata Institute of Fundamental Research

In 1970's the Tata Institute of Fundamental research (TRFR) library developed a program for making monthly list of new additions to the library with a keyword index. The TIRF library developed new systems for stock verification using computers which made the job of stock checking more accurate, easy, time saving and regular. The TRFR library receives a huge number of reports and preprints. They have also developed a system generate a weekly list of documents with index for report number and personal author.

6) Bhabha Atomic Research Center (BARC)

The BARC continues to do a lot of better work in providing Selective Dissemination of Information (SDI) service using INIS tapes with MISTRAL software. Apart from this the center BARC developed a system 'Automation for Storage and Retrieval of Information' (AFSART). BARC become one of the first institutes in India to setup a CD-ROM facility, when it installed a Philips in 1988.

7) Physical Research Laboratory (PRL)

The PRL library Ahmadabad is the initial stages started issuing a computer based Library additions list with a KWOC index. These programs were written in PL/1 and SPS languages and were run on IBM 360/44 and IBM 1620 computer.

8) Periodical Management System (PMS)

A PMS was developed by Ghiya, Bharucha and Kulkarni in 1980. This system understands all activities connected with dissemination of information and procurement recording. This program was written in FORTRAN IV and in is run on IBM 360/44 computer.

9) Book Procurement System

Bakre, Kulkarni, and Bharucha developed a computerized Book and Procurement System (BPB) at PRL, Ahmadabad in year 1981. This system observer the status of processing of every one individual request received by the library and later assists the library staff in taking quick action.

10) Indian Institute of Technology, Delhi

Haravu developed a computerized acquisition routine of serials in 1969 at Indian Institute of Technology, Delhi. Initially, eighty five periodicals were selected for this experiment on the basis of country, subject and frequency of origin.

11) Indian Institute of Technology, Kanpur

In fact, first experiment in computerization was conducted at IIT Kanpur in 1964. The INSDOC had distinct to use the IBM 1620 computer at IIT, Kanpur for the compilation of the Union Catalogue in that year.

12) Indian Institute of Technology, Madras

Use of IBM TEXT-PAC: Indian Institute of Technology, Madras purchases the IBM 370/155 computer in 1970s. Since then number of experiments and operation concerning to library automation and bibliographic data services were carried put doing well. Ramanujan and other conducted a study of IBM TEXT-PAC software packages at IIT, madras. At a huge stage, INSDOC had started the SDI services from the same center using CAN/SDI software packages. The IIT Madras library too did some task in library automation.

13) Bharat Heavy Electricals Ltd. (BHEL)

The BHEL R and D unit, Hyderabad decided a computerized book ordering system on 1982. The software was purchase from the Clark Library System (CLAS) of USA and modified to suit the required of BHEL library.

14) National Aeronautical Laboratory (NAL)

The NAL library at Bangalore has been using electronic data processing software and hardware for book acquisition and ordering since long time. The NAL has also automated a circulation control system.

15) Hindustan Aeronautical Ltd. (HAL)

Kumar developed uncomplicated program in COBOL to automate the circulation system of Hindustan Aeronautical Ltd. (HAL), Bangalore.

16) IDL Chemicals Ltd.

The library of the IDL Chemicals Ltd. Hyderabad has been using a computer for different library activities science 1980. All the programs are being run on their V-76 minicomputer system.

17) Central Mechanical Engineering Research Institute (CMERI)

The CMERI, Durgapur stared automation of library services using the IBM 1620 computer available at the institute. They had written several programs for providing SDI service to their users.

18) Electronics Commission (EC)

The Electronics Commission (EC), New Delhi founded Information Planning and Analysis Group (1GAP) in 1971. This has developed a database for books on the HP21 minicomputer systems. It uses IMAGE/2000 database management system.

19) Institute of Armament Technology

The defense Research and Development organization (DRDO) has established forty three laboratories and field units spread through in India. It was establish an information network liking different R and D units of DRDO keeping in view of requirements of such a system the Institute of Armament Technology (IAT) library, Pune initiated certain experiments in computerized information processing.

20) National Physical and Oceanography Laboratory (NPOL)

The National Physical and Oceanography Laboratory (NPOL) library at Cochin developed a computer based periodical control system using TEKTRONIX 4051 desk computer model. The original journal by journal record in the Kardex file was transferred to the computer memory.

21) Punjab Agricultural University (PAU)

The PAU, Ludhiana was using Friden Flexo writer for the reproduction of catalogue cards and for making masters for the offset machine.

22) Regional Research Laboratory (RRL)

Regional Research Laboratory (RRL), Hyderabad Raoof and Prasad developed of system of computerized reminder for non receipt of periodicals. The program for this was written using FORTRAN IV and run on the laboratories IBM 1620 computers.

23) Defense Scientific Documentation Center (DESIDOC)

DECIDOC has developed an online interactive information storage and retrieval system for processing bibliographical information. The requisite information as well as software technology was developed in house. Now CD-ROMs can be used as a substitute to online searching and even to build up customized databases. **(S. Jageesh 1998).**

4.19 CHARACTERISTICS OF AUTOMATION

The important characteristics of an automated system are follows:

1) The processes/ operations are carried out automatically.
2) Reduced or avoid human action and thus save labor and time.
3) It speed up efficiency and speed in operations.
4) Spends the range and raises quality of existing services.
5) Immediate answer to multiple quarries.

4.20 NEED OF LIBRARY AUTOMATION

A Computerized library is always best than a manually operated library in terms of processing, organizing and dissemination of information to the users efficiently and affectivity. If a library automated and connected to various library networks such as DELNET and Information Library Network (INFLIBNET). It provides an opportunity to the users i.e. research scholars, students and teachers to have and access to the academic world. A library whether academic, national, public or specialist main functions and objectives are to collect disseminates information to its users efficiently and effectively application of Information Technology in the organization and services of libraries have brought about sea changes in the functioning of libraries in the last 2 decades throughout the world.

Almost all the advanced libraries are using computers for their daily library operation in addition to housekeeping activities and some the big and modern libraries are using the computer network using communication satellites for resource sharing use which enables the participating library to acquired not only bibliographic details from other library's collection but also fullest abstracts and indexes of the required articles using Local Area Network (LAN), fax, email and internet attachment to the system library automation involving creation of database and information retrieval, use of telecommunication for information handling and computerized library network, require careful and systematic planning. (**Singh Garudev 2007).**

4.21 NETWORKING

Technology nowadays is moving very speedily and in this age of electronic revaluation and computers, more and better gadgets are being invented each and every day which make our lives easier and simpler. Communication too has developed tremendously due to the advances made in communication technology such as blue-tooth and WAP technology we are moving towards a seamless environment where every forms of communication are merging.

The concept of networking is in all likelihood as old as telecommunication itself. Consider people living in the Stone Age, when drums may have been utilized to transmit message between individuals. Certainly, we have come a long way off from the primitive towards and devices of our forebears, in the present, we have computers talk to each other over wide assemblages of wires, microwaves and fiber optics.

The simple meaning of a computer network is, any set computers or devices attached to each other, However, we define a computer networks as a collection of host that are capable to communicate with each other using a single technology, frequently by relying on the services of a number of dedicated host that transfer data between the participants. Hosts are many computers, but one can also think X terminals or clever printers as host. Small collections of host are also called sites.

The latest development in the network user population has brought in to the arena huge number of non-technical applications, such as e- mail, and wish to explore the growing availability of network information resource. The growing complexity of the network environment, together with the increase in user access as physical computer, brings into sharper focus; require for easy to use of access tools and navigational information discovery. Information can sometimes be available speedily in electronic form now than in the printed equivalent.

The importance is therefore moving away from the conjectural and uncertain implementation of technical characteristic and towards the design of reliable network services, which are suitable for direct use by a non specialist without experience, training and support. A clear trend is towards a situation in which the end or last user interacts directly with the information source, quite than through a human intermediary. Plenty of the problems in organization and making reachable this confusing wealth of information

resources seem comparable to those which the traditional librarian has faced with printed material, while the solution may not be straightly transferable.

One area in which librarian has been energetic in the compilation of resource guides to network provisions; this has been a specific feature of the research networks environment. The online text document or printed guide is not well suited to inclusive coverage of the quick changing network environment. A database which can be essential from any network node can provide more authoritative and faster information on changing resource and location. (**Singh, Manvendra 2006).**

4.22 TYPES OF NETWORK

1. LAN (Local Area Network)

In a Local Area Network (LAN), two or more computers are directly linked within a small well define area such as office building, campus and room. It has relatively few numbers of users. They are comprehensively used to connect personal computers and work station in factories and company offices to exchange information and share resources. These are distinguished from other types of networks by the three characteristics (I) their size (II) topology (III) transmission technology.

LAN links a number of computers (work station) together to permit plenty of people to utilize the same computer programs and share information. Each and every user can access the network form an individual computer work station. It provides multiuser environment which permitted sharing of data as well as expensive resources like printer and storage. It also provides connectivity, modularity, security superior performance and reliability in its operation.

It can suitable any site requirement and in it any number of users can be tailored to suitable any type of application. Main advantage of LAN is the reduction of software and hardware costs because users can share several computers, secondary devices such as LAN platform software, modem, hard disks and color plotters. Another advantage is that the users can share the same data. **(Mandal, Datta & Podder 1998).**

2. MAN (Metropolitan Area Network)

The MAN network is basically a larger version of LAN. It is called metropolitan since it generally covers the area of a large city. It might be either public or private.

Different transmission media and hardware are often used in MANs because they must efficiently cover these distances up to tens of kilometers. A MAN can support both voice and data, and might even be connected to the local cable television network. A MAN has just one or two cable and does not include switching elements, which shunt packets over one of the some potential output lines.

3. WAN (Wide Area Network)

A WAN is composed of a number of autonomous computers that are distributed over a huge geographical area, often a continent or country. It can be implemented with the support of public networks as well as private networks. Private networks are developing within a one corporation. The implementers lease circuits for private use which are normally telephone lines and construct a network. On the other hand, public networks are developing by government telecommunication agencies. The switching and transmission facilities are shared by several corporation and organizations.

The majority of WANs, the network contains numerous telephone lines or cable. It has various uses and it makes file transfer simply. They are useful in data collection. In addition to these, WANs have plenty of other users, including software distribution, the capacity to use expensive resource, such as high resolution color graphic printers and computer servers, importantly and remotely, the capacity to manage networks centrally. Wide networks are normally limited to use by the large government agencies and organization due to high costs involved in maintaining and building them. It is a satellite based wide area network with national coverage. For example: MALIBNET, PUNENET, BALINET,

4.23 HARDWARE AND SOFTWARE REQUIREMENTS OF LIBRARY NETWORKING

A) Hardware Requirements

All of the member libraries must have at least some of the following hardware. However, the central library (Host) may need most of the hardware mentioned below, but with additional disk space. Hardware specifications are given below:

1. Server:

i. 1.2 GB HDD

ii. Pentium @ 233 MHz with 64 MB RAM
iii. Windows-NT Operation System
iv. Color Monitor (SVGA)
v. 32 x CDROM Drive
vi. MS-SQL Server 6.5

2. Client:

i. 1.2 GB HDD
ii. Pentium @ 233 MHz with 64 MB RAM
iii. Windows-98 Operating System
iv. Color Monitor (SVGA)
v. 1.44" Floppy Drive

3. Other Hardware:

i. Modem (With dedicated phone line)
ii. Fax (With dedicated phone line)
iii. Ai lest two determine printers
iv. DD-ROM drive
v. Cartridge tape drive
vi. One letter quality printer (preferably laser printer)

B) Software Requirements

All of the participating libraries must have the following software need:

i. Software for effective network operation as well as the library automation
ii. UNIX (latest version, to begin with, it may be optional)
iii. DOS (latest version)
iv. World processors
v. LAN (Novel/Ethernet, to begin with, it may be optional)
vi. DBMS packages
vii. Software for E-mail (at list PROCOMM)
viii. IR packages CDS/ISIS
ix. Programming languages : C/C++
x. Library automation: Granthlaya / Libsys / any other similar packages.

4.24 COMPUTER NETWORKING IN LIBRARY

It is nothing but connection between two or more computers systems through cables. It is required for sharing the information between the different systems. Computer networks are very important for library modernization. The Public Switched Telephone Network (PSTN) is computer-controlled. There is a lot of scope for communication through computer network system. Due to this there is a lot of demand for computer hardware, software and peripherals industries.

4.25 LIBRARY NETWORK

The recent growth in the network user's population has brought in to the arena larger numbers of non-technical people who have learned the advantages of creation network applications, such as electronic mail and wish to explore the increasing availability of networked information resources. The increasing complexity of the growth in user access as physical computer brings into sharper focus the need for easy to use navigational information discovery and access tools. Information can sometimes be available faster in electronic form now than in the printed equivalent.

The emphasis is therefore moving away from the speculative and uncertain implementation of technical features and towards the design of reliable network services, which are appropriate for direct use by a non-specialist without extensive support and training. A clear trend is towards a situation in which the end user interacts directly with the information source, rather than through a human intermediary. Many of the problems in organizing and making accessible this confusing welt of information resources seem analogous to those which the traditional librarian with printed material, although the solutions may not be directly transferable.

Although such tools have begun to emerge, they are at present developing from technical and computing tend to lack a sophisticated view of the information transfer process. In this concept some of the main information discovery and retrieval tools are discussed as examples of these emerging trends it is clear however that these are merely the ancestors of future approaches which will more truly address the underlying problems.

The main arena in which these type of tools have been in the field of research and academic networking since network resources are seen as one form of educational resources to be exploited for both research and learning purposes and perhaps because an opportunity has existed in this environment to develop such tools in response to perceived need without early commercial pressures.

One area in which librarians have been active has been in the complication of resource guides to network facilities this has been a particular feature of the research networks environment. The oriented guide or online text document s no well suited to comprehensive converge of the fast changing network environment. A database which can be queried from any network node can change resources and location. The scale of the problem is such that no single centralized and updated guide is likely to be feasible in providing these mechanisms.

A resource guide can consist of a central database which can be updated from local sources with local information or a distributed database which can redirect queries to the local site for processing in a web of co-operating databases. A distributed X.500 directory would seem an attractive option here, although there is doubt whether the present standard is capable of providing an adequate solution at the scale which the problem requires.

The distributed approach presupposes a level of agreement on standard descriptions for classes of item and on protocols for the routing of queries between local databases. To provide the user with immediate access to the resource the guide would need to contain routing information as well as a description so that sought items could be retrieved in full without further action by the users.

Several proposals have been made on schemes for organizing and uniquely identifying resources particularly in the internet world. There are however a number of problems with electronic information item in such an uncontrolled environment the possibility of loss of information in different versions and at content between different instances of the same record the problem of version control the identification of name changes in for example filenames for items with the same content.

The use of MARC format cataloguing records has been proposed as a more traditional solution and the feasibility of this approach has been investigated by the

OCLC organization. The final report proposed the creation of a new field for location and access information but suggested that the relative scarcity of formally published information made the extension less attractive. No clear scheme has emerged as having majority agreement and it may be that a traditional universal citation and mapping approach is not an appropriate solution. Most attention at the present time is focused on the creation of directory tools.

The Top node project is an attempt to create an overall directory of directories, catalogues, resource lists, network services and tools together pointers to other sources of information. Top node has developed a proposed record format which includes pathway information for locating an item as the structure for a centralized database which will update local copies of the data. A newly formed Internet Network Information Centers proposes to develop a similar directory of directories and maintain resource description files with regular updates. This will be a distributed information base and will be able to dispense resource information in discipline specific packets. The development of internet as a National Research and Educational Network and the growing emphasis on networking for economic development seems likely to encourage these more formal initiatives.

Hytelnet is a terminate and stay resident hypertext browser which provides an IBM-PC user with information on access to library catalogues, campus-wide information system, free-nets, bulletin boards and other network resource tools which may be contacted through the internet. The program gives information on using the Telnet IP protocol to access these systems but does not make the connection itself. Despite this it is widely available in the public domain and is updated regularly. The package is a useful adjunct to the desktop workstation user, acting as an aide-memory rather than masking the complexities of internet communications.

Online directories which do exist have much more limited aims and cover only small, specific areas. The WHOIS sever is presented on many internet sites, containing information on network users on a limited scale, by offering directory services related to the organization to which it belongs there is no attempt to share a common directory with other organizations. Servers of this type however can be accessed from other points on the network either by remote log in or through electronic mail. Bulletin boards and

campus information systems may also contain directory information of this short on a rather ad hoc basis and with often unsophisticated searching mechanisms.

Much of the work on more structured information discovery and information management tools has again emerged from the internet, reflecting the great amount of freely available information and a user community with the interest and skills to develop the tools. For this reason such tools are designed to run across TCP/IP based networks and normally demand a direct connection from a host system or for the user to have remote access to one.

They have usually developed from research project and many are in the public domain rather than being commercially available as products although there is a tendency for the most successful to move in this direction. Information discovery tools are software programs which perform specialized task for one particular kind of resource: retrieval of text documents identification of software programs or management of user access. They are designed to perform two main interrelated tasks to manage and make available the contents of large databases and disk directories from one location and to assist a remote user in finding and accessing information from a number of different sources at different locations.

The tools described here use a client-server model to achieve these two functions. The server is responsible for maintenance and management of the information base dispensing answers to queries submitted by a client. The client is responsible for interacting with the user in the most appropriate way managing the user interface and passing queries back to the server in an agreed format without requiring the user to know what this is. A protocol binds these two parts together forming the common language into which both queries and results are coded foe transmission across the network. Since these tools have evolved from the internet environment, purpose built protocols of this sort are predicated on the use of TCP/IP as the underlying transmission medium. **(Sing Manvendra, 2006)**

4.26 NETWORKING OF PUBLIC LIBRARIES

With computerized data access through internet, all types of information like social security, legal, medical aid, travel schedule etc. can be collected. The public

libraries foreign countries are making full use of it. Since libraries in India have not been a priority development is not much. Public libraries could serve in improving literacy but suffer due to lack of funds, modern equipment, good and adequate collection, proper reading rooms, trained staff and user facilities. No doubt in today's world public libraries have to be technologically advanced so that they have access to INTERNET and international database and network to serve the people better.

According to UNESCO public library Manifesto 1994:

"The public library acts as a living force for education, culture and information and as an essential agent for fostering the peace and spiritual welfare through the minds of men and women"

The Manifesto recommends that "The Public library network must be designed according to national and regional research and special libraries as well as libraries in schools, colleges and universities should be considered."

Accordingly there is a need to establish a public library network which is in compliance with the other networks in the country. There are around 60,000 libraries in the country. They can't be considered as libraries in the real sense as most of them provide only reading facility like newspapers, magazines or story books etc. Most of them even do not have computers. Training of library staff in the computer application is also important for smooth functioning of library operations. The settlement in rural area is generally spread over in remote places where facilities like electricity, road is not available so there is a scarcity of trained staff.

As the infrastructure is not suitable and the people are not so educated it is difficult to have network between rural libraries and public libraries. But due to wireless communication technology it has become possible to join the network of other libraries. This way Public library can reach the public get connected with the national network system. We have to develop the infrastructure of public libraries. We have to improve manpower by training them in AACR-II, MACC format, Library Congress subject headings, handling of software and use of electronic mail and INTERNET etc.

4.27 NETWORKING RESOURCES IN PUBLIC LIBRARIES

The networking system is serves the purpose in two ways. One is that it provides educational support to the needy people of the society and the other is a facility for continuing education for the working class and left outs that have difficulties in getting education. Thus public library system is a very important and essential part of the state. Its job is to maintain collection of books and other reading material and serve the public free of charge. This way the public literacy will increase. If we look back we find that even after so many years of independence our nation has very poor system of public libraries. States having formal library legislations are also not have proper library system though people are paying a library tax. The state budget and grants are too small to keep up public library services going and their growth and modernization. **(Iyer V. K. 1999).**

4.28 AIMS AND OBJECTIVES OF LIBRARY NETWORKS

The main and basic purpose for developing a network is to provide information services to group of libraries through sharing of resources of the involved libraries of the network. This may guide to member libraries to depend additional on access to document held in the other member libraries than on depending only on their respective collection. The important and main aims and objective of library network are stated as the following.

1. To upgrade resources utilization and service level at the particular libraries by providing automation provision in the following areas
    a) Fund accounting and acquisition.
    b) Circulation
    c) Serial control
    d) Cataloguing - assist member libraries in cataloguing of serials, books, catalogue production and non-book materials
    e) User services – implementation electronic services in the libraries for speedy communication of information
2. To providing resource sharing and co-operation activities between libraries by providing reliable and efficient resource sharing, for example;
    a) Document Delivery Services – providing the copies of the document which is not accessible in their respective libraries

b) Inter library loan (ILL) for maximum use of resources
c) Access to international and national database
d) Refresher course facilities and manpower training – stimulating, promoting and co-ordination research and training programmed for network members and library staff
e) Communication link through publication and inter – personal communication

3. To started referral centers to monitor and maintain a central on-line union catalogue of serials, books, non-book materials and facilitate catalogue search of all the participating libraries.
4. To coordinate efforts for appropriate collection development and reduce unneeded duplication wherever possible. The purpose is
   a) To reduce the operational cost of the collection development
   b) To develop collection by mutual cooperation
   c) To control the rate if easing cost of the collection development
5. To create a specialist bibliographic database of non-book materials, books and serials for search and access.
6. To coordinate with other international, national and regional networks for exchange of documents and information for the use of libraries and user.
7. To develop a database of specialists, project and institution for providing online information services.
8. To evolve standards and uniform guidelines in method, techniques, procedures, software and hardware services and so on and promote adoption in actual practice by every libraries, in order to facilities towards optimization and facilitate pooling, sharing and exchanging resources.

4.29 FUNCTION OF LIBRARY NETWORKS

At some places library networks have been established for sharing of resources due to unnecessary wastage of finance can be avoided. Sometimes all the networks do not conform to the essential function of other libraries. The essential functions should include the promotion of resource sharing, union catalogues, and systematic entries of acquisitions and implementation of international standards for making of records

uniformly. Considering this the functions of a typical library networks might fall into the following three categories.

A) Information Services

1. Inter library loan service – it means that every member library may need having the facility to photocopy for delivering the documents.
2. Reference and referral service – every member library may need the facility of having dedicated E-mail services or/ and telephone.
3. Access to databases – to prepare backdated bibliography for the purpose of browsing to know whether or not a document is accessible.

B) Management Services

The management services i.e. function that supports the network administration are:

1. Evaluation of the networks, this is done through
    a) Collection of statistics
    b) Analysis of performance of the network
    c) User studies
2. Establish an operational system that implement the function mentioned above
3. Training activities. These are done through
    a) Staff development programs
    b) User oriented programs
4. Communication of activities such as
    a) Holding of meetings. etc
    b) Publication of newsletter
5. Cost analysis. It involves
    a) Fess to be collected, determination of cost, etc.
    b) Allocation of budgets

C) Technical Services

Function that serve the libraries i.e. the technical services are:

1. Technical processing involved in acquisition
2. Co-operative collection development program
3. Circulation control system

4. Cataloguing and other means of resources to identify and to locate documents. **(Gahale P.D. 2013)**

4.30 NEED FOR LIBRARY NETWORKING

1. The problem is the increase prices of publications, which has pretentious collection development in libraries.
2. The ratio of increase of knowledge and information is faster than before and it is ever increasing. Therefore, it has become not possible for all libraries to acquire every document that is published in the library.
3. Due to the emergence of recently developed subjects, readers need pin-pointed and exact information that may be available in other libraries.
4. The budget and funds of the library is not raise and this makes it complicated for the individual library to provide services from its individual collection.

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